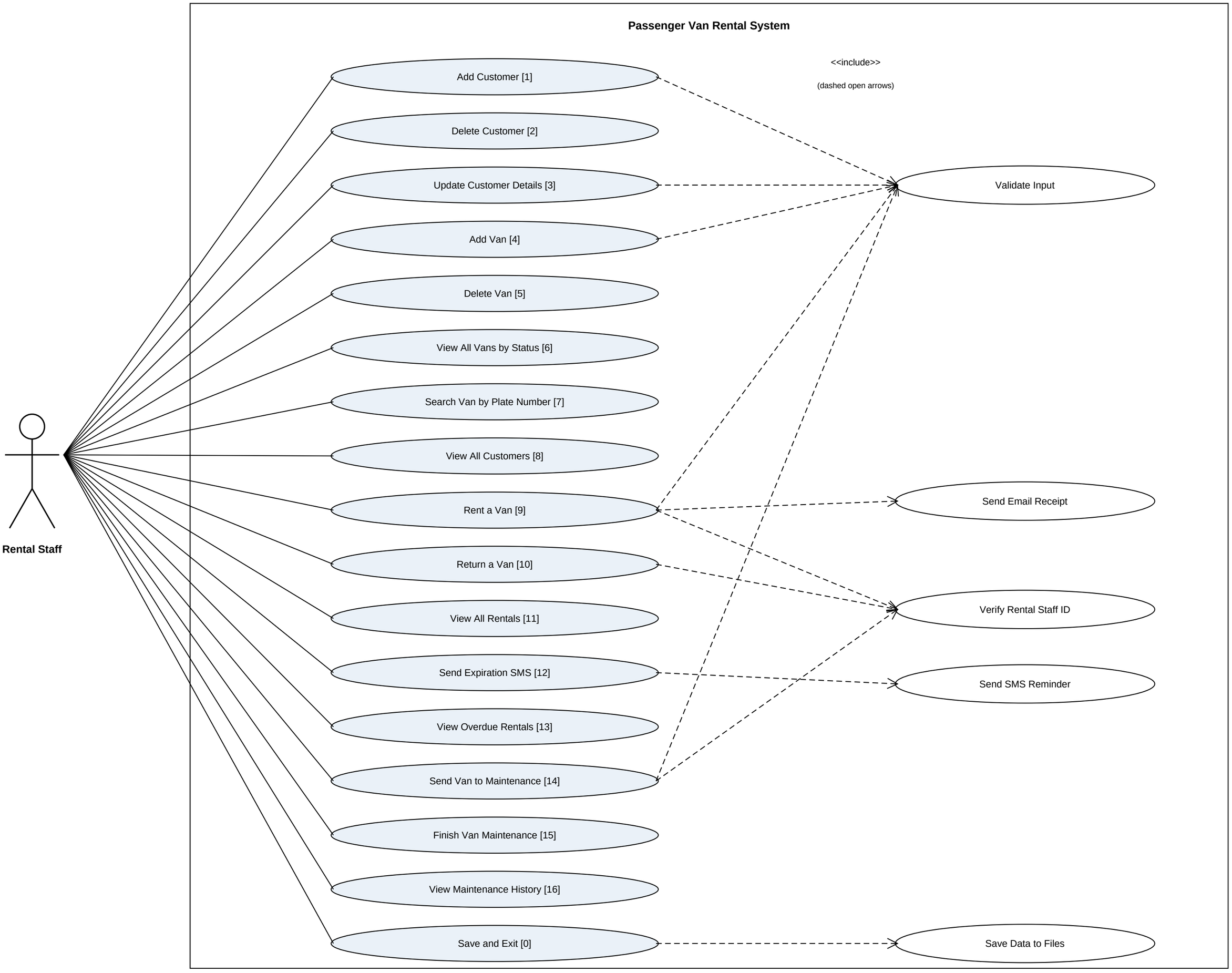
1 Use Case Diagram

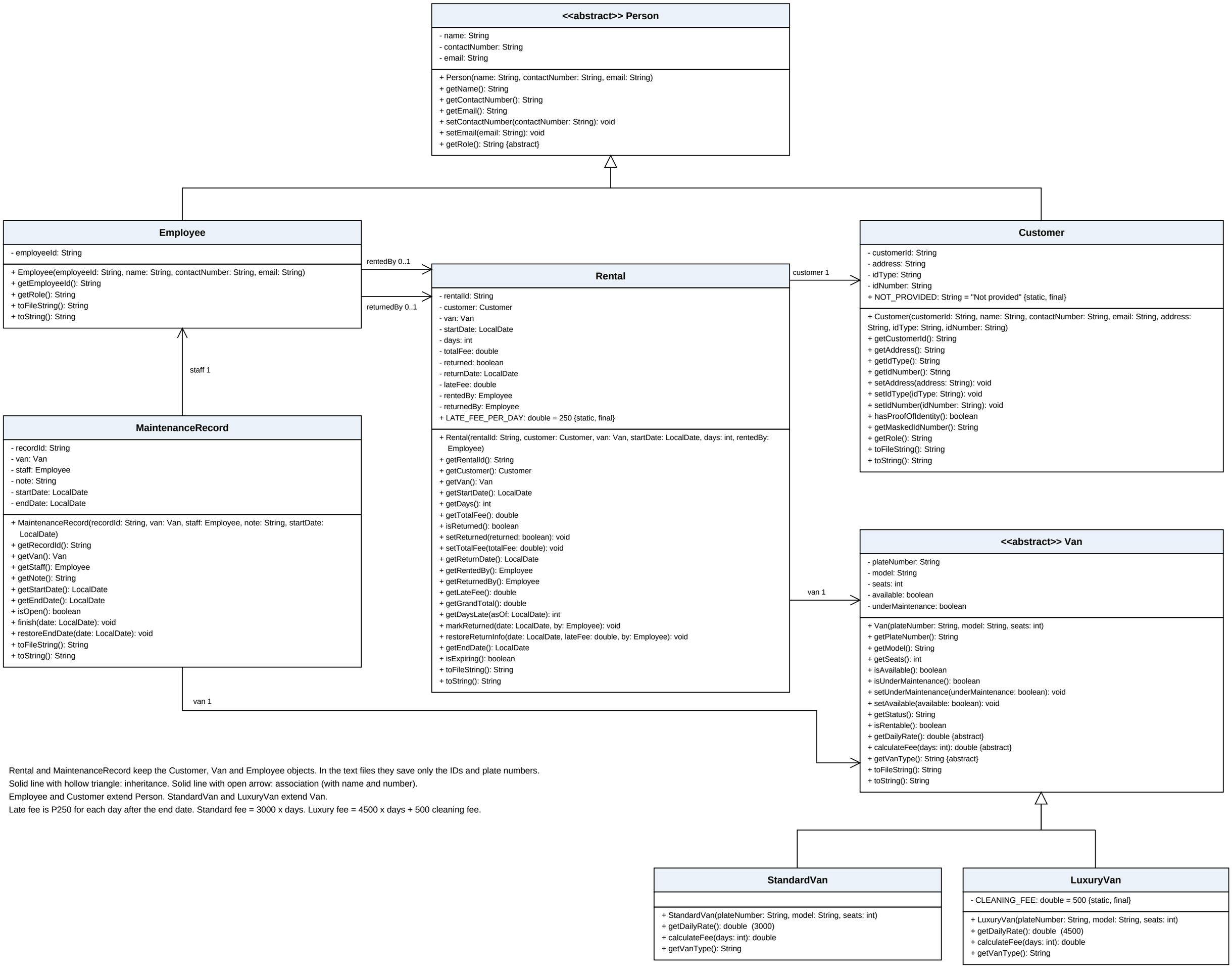
Console app with menu options 1 to 16 and 0. There is no login, so there is only one user: Rental Staff.



Menu numbers match the console menu in Main. Add Van asks for type S (Standard) or L (Luxury).
Rent, Return and Send to Maintenance ask for a Rental Staff ID first. The ID is saved in the record.
A customer needs an address and a valid ID before renting. A customer with a rental record cannot be deleted.
Data is saved after every menu action, not only on option 0. Email and SMS are not real: they are printed and saved in notifications_log.txt.

2 Class Diagram: Domain Classes

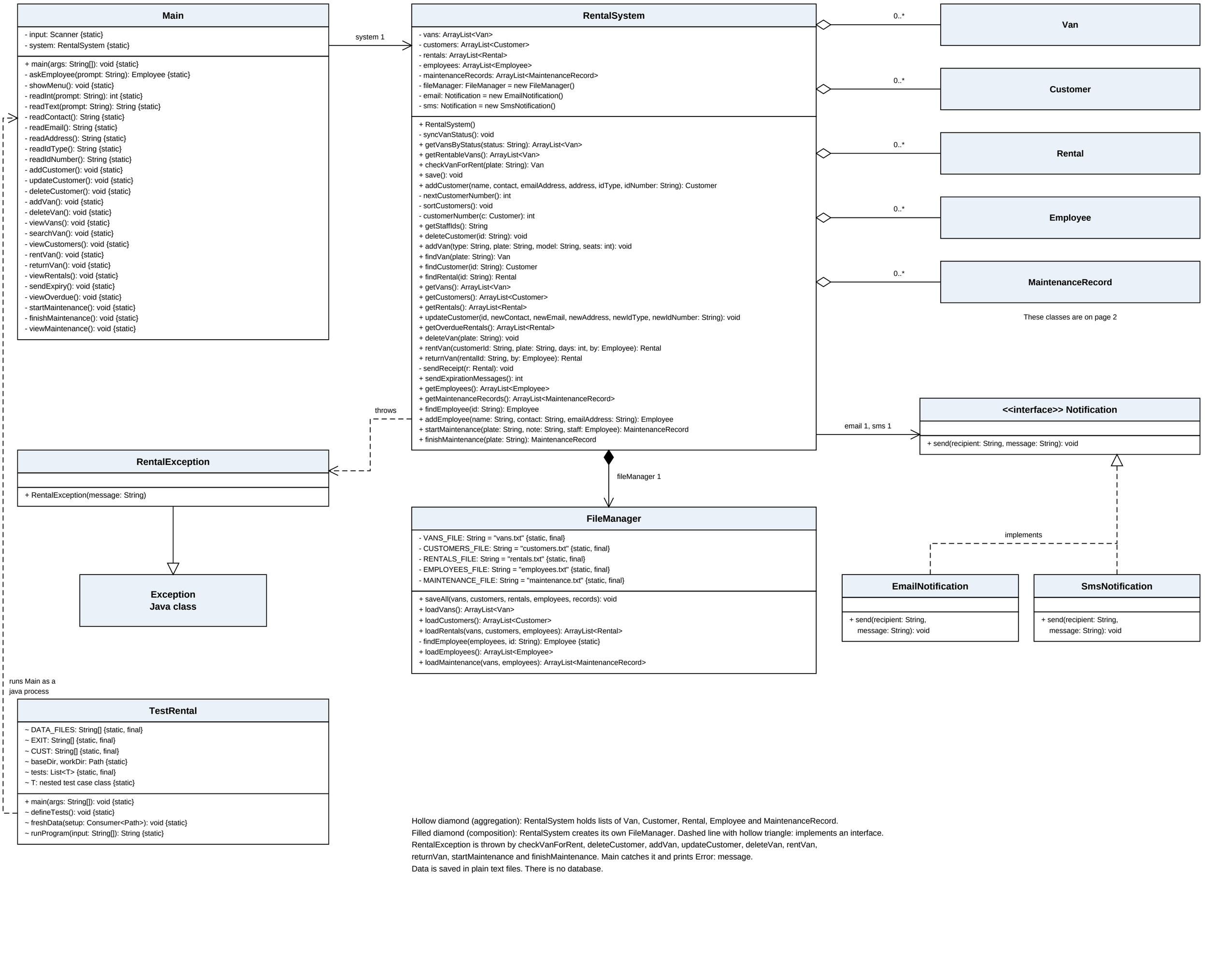
Fields and methods match the Java files. {static} and {final} are Java keywords. The other classes are on the next page.



Rental and MaintenanceRecord keep the Customer, Van and Employee objects. In the text files they save only the IDs and plate numbers.
Solid line with hollow triangle: inheritance. Solid line with open arrow: association (with name and number).
Employee and Customer extend Person. StandardVan and LuxuryVan extend Van.
Late fee is P250 for each day after the end date. Standard fee = 3000 x days. Luxury fee = 4500 x days + 500 cleaning fee.

3 Class Diagram: System, Services and Support Classes

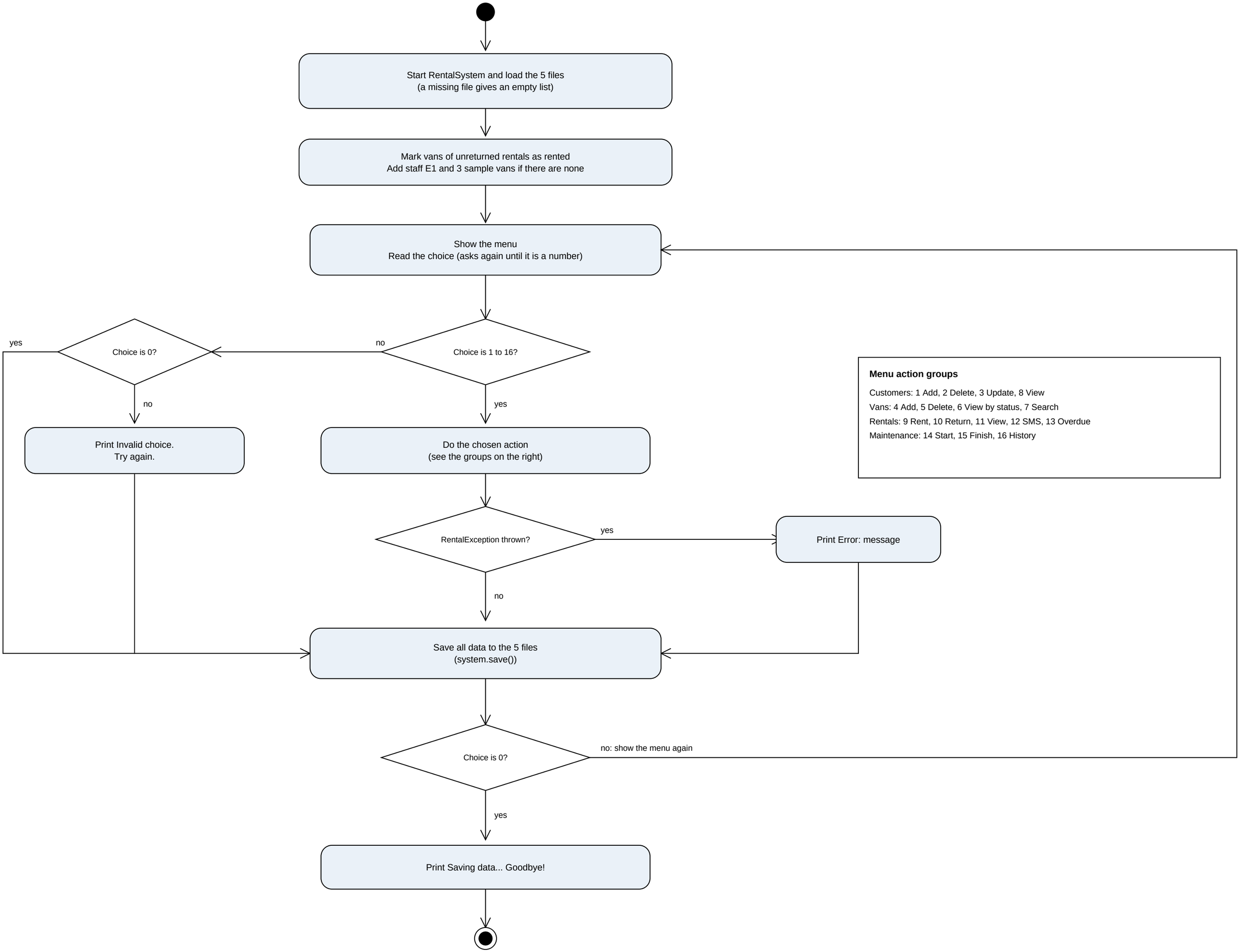
Main handles the screen and keyboard. RentalSystem keeps the lists and the rules. FileManager saves and loads the text files.



Hollow diamond (aggregation): RentalSystem holds lists of Van, Customer, Rental, Employee and MaintenanceRecord.
Filled diamond (composition): RentalSystem creates its own FileManager. Dashed line with hollow triangle: implements an interface.
RentalException is thrown by checkVanForRent, deleteCustomer, addVan, updateCustomer, deleteVan, rentVan, returnVan, startMaintenance and finishMaintenance. Main catches it and prints Error: message.
Data is saved in plain text files. There is no database.

4 Overall Activity Diagram

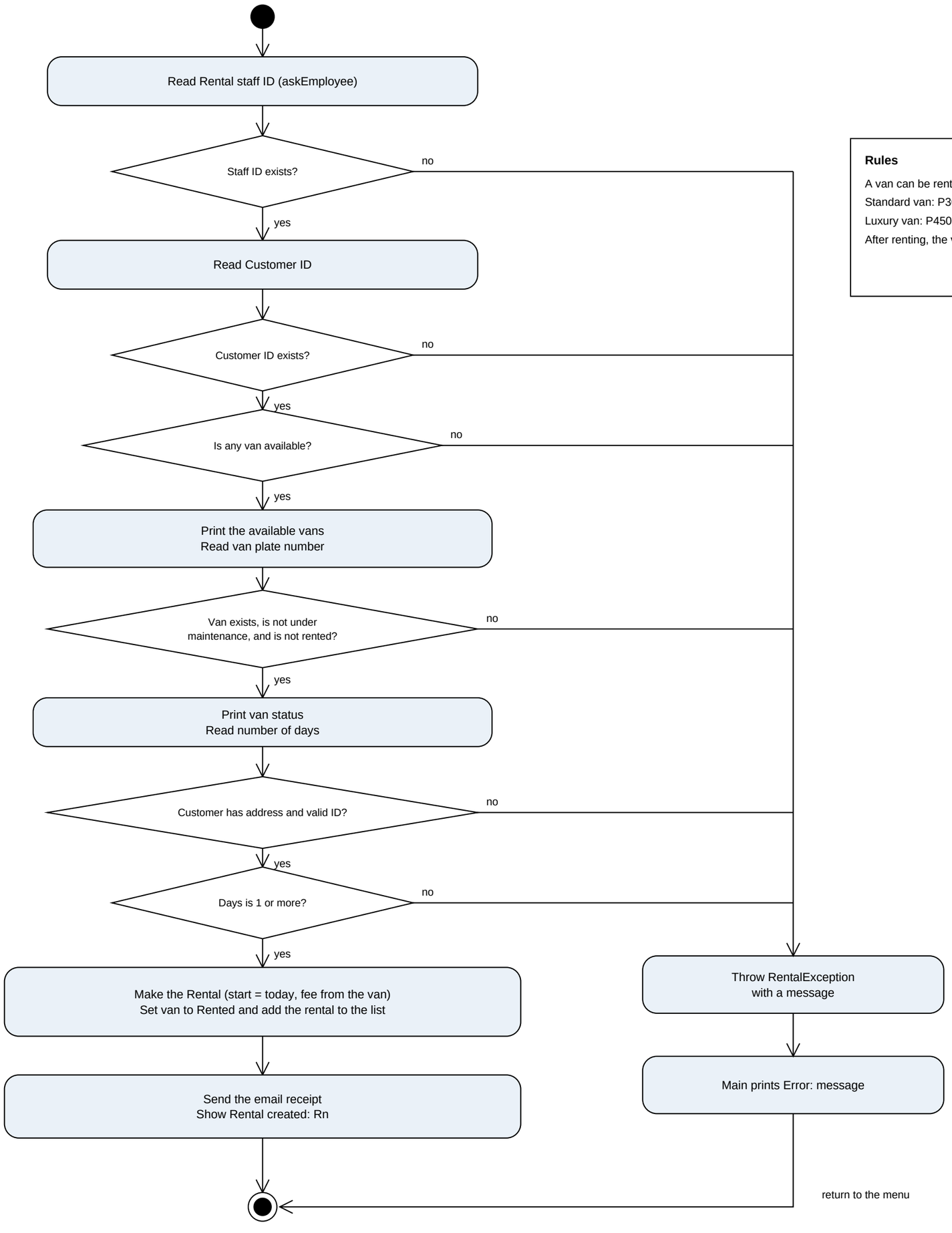
The menu loop in Main. Data is saved after every action, even after an error or a wrong choice.



Actions 9, 10 and 14 ask for a Rental Staff ID first. Input is asked again until it is valid (a number, no commas, contact starts with 09, email format).
An error from a broken rule does not stop the program. The error is shown, the data is saved, and the menu comes back.

5 Rent a Van Activity Diagram

Menu option 9. If a check fails, an error is shown and no rental is made.



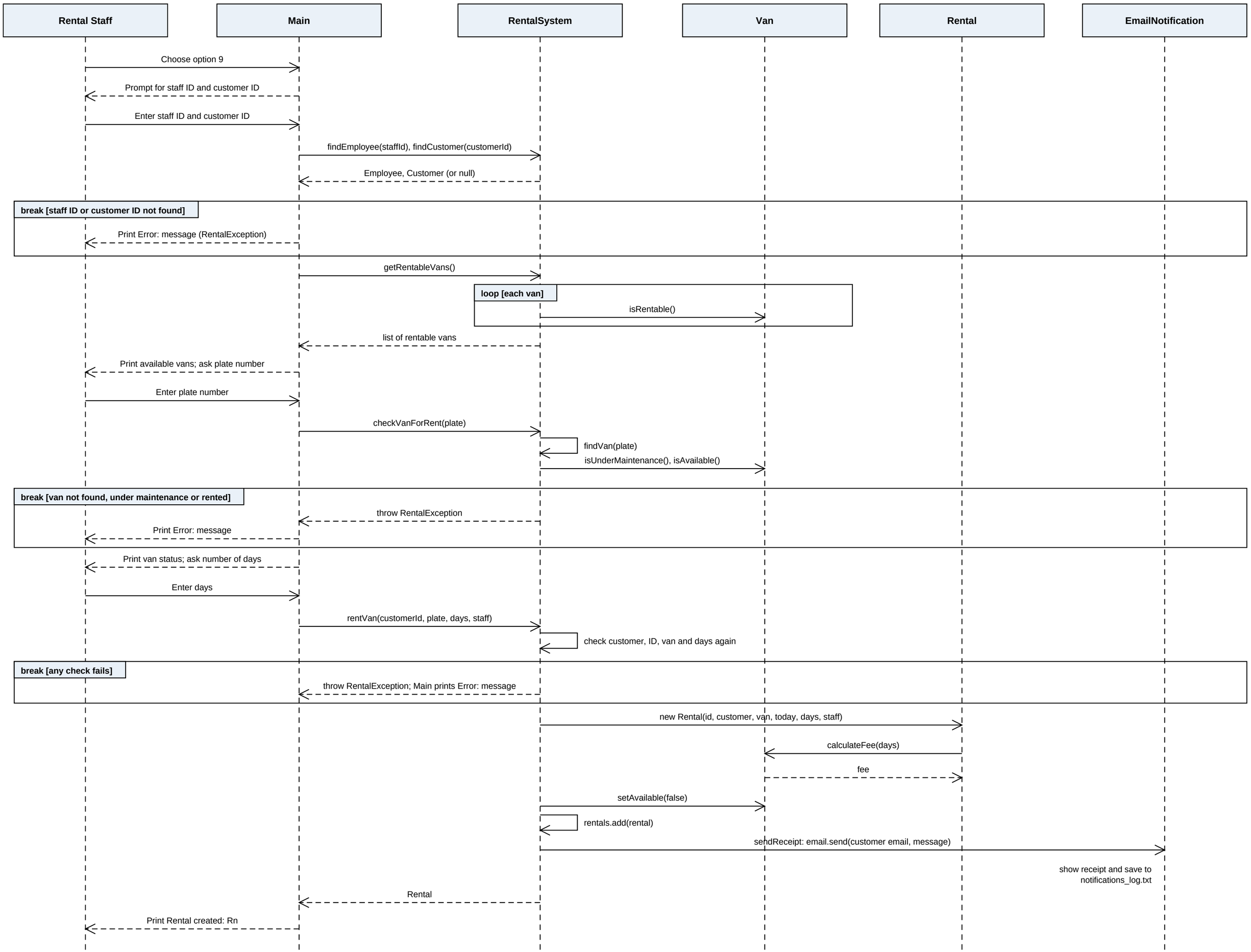
Rules

A van can be rented if it is Available and not under maintenance.
Standard van: P3000 per day.
Luxury van: P4500 per day + P500 cleaning fee.
After renting, the van status is Rented.

Main checks the van first to show its status. rentVan checks again. The rental ID is R + (number of rentals + 1).

6 Rent a Van Sequence Diagram

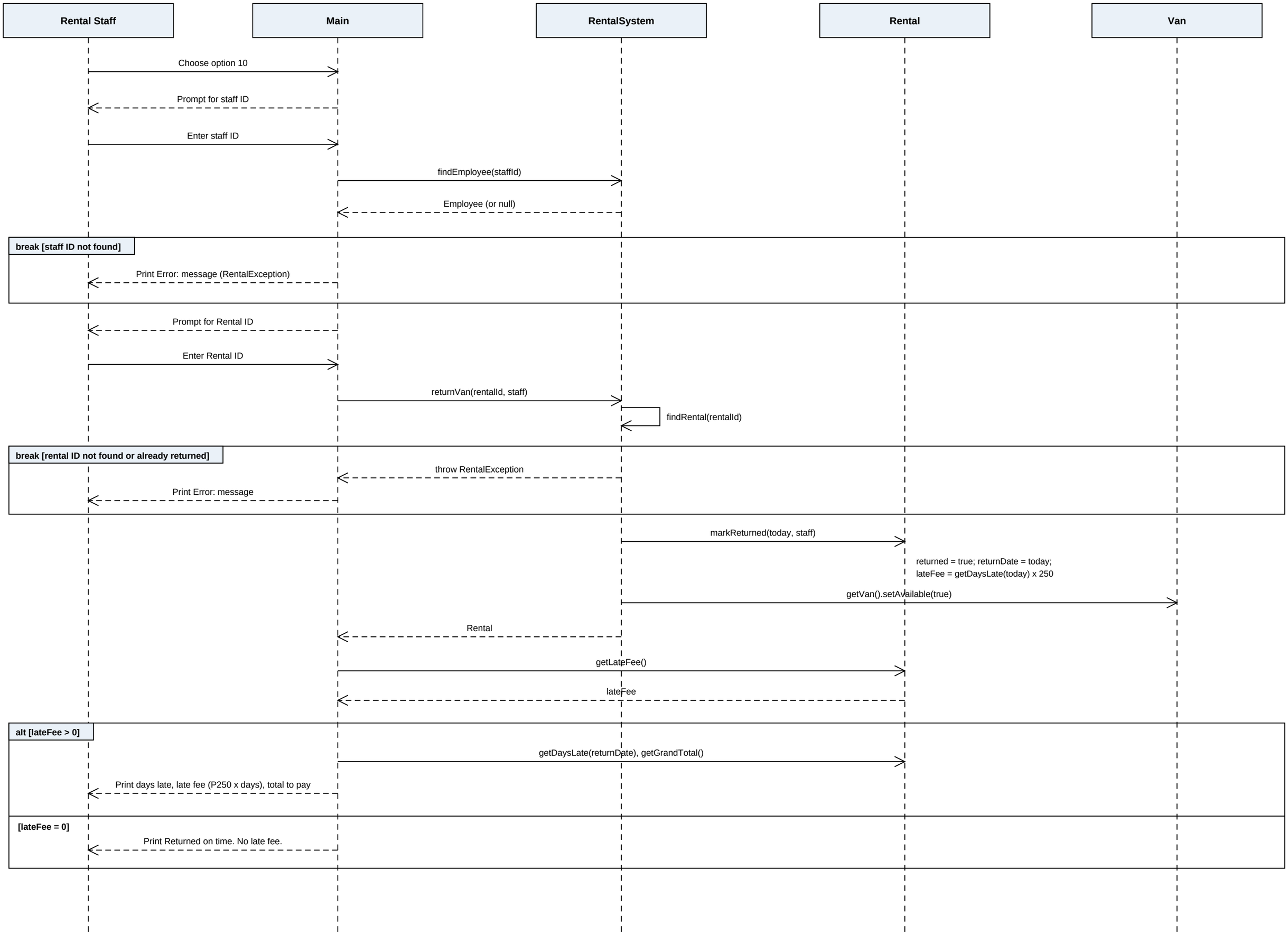
Menu option 9. If a check fails, it goes back to the menu and no rental is made.



After the action, the main loop calls system.save().

7 Return a Van Sequence Diagram

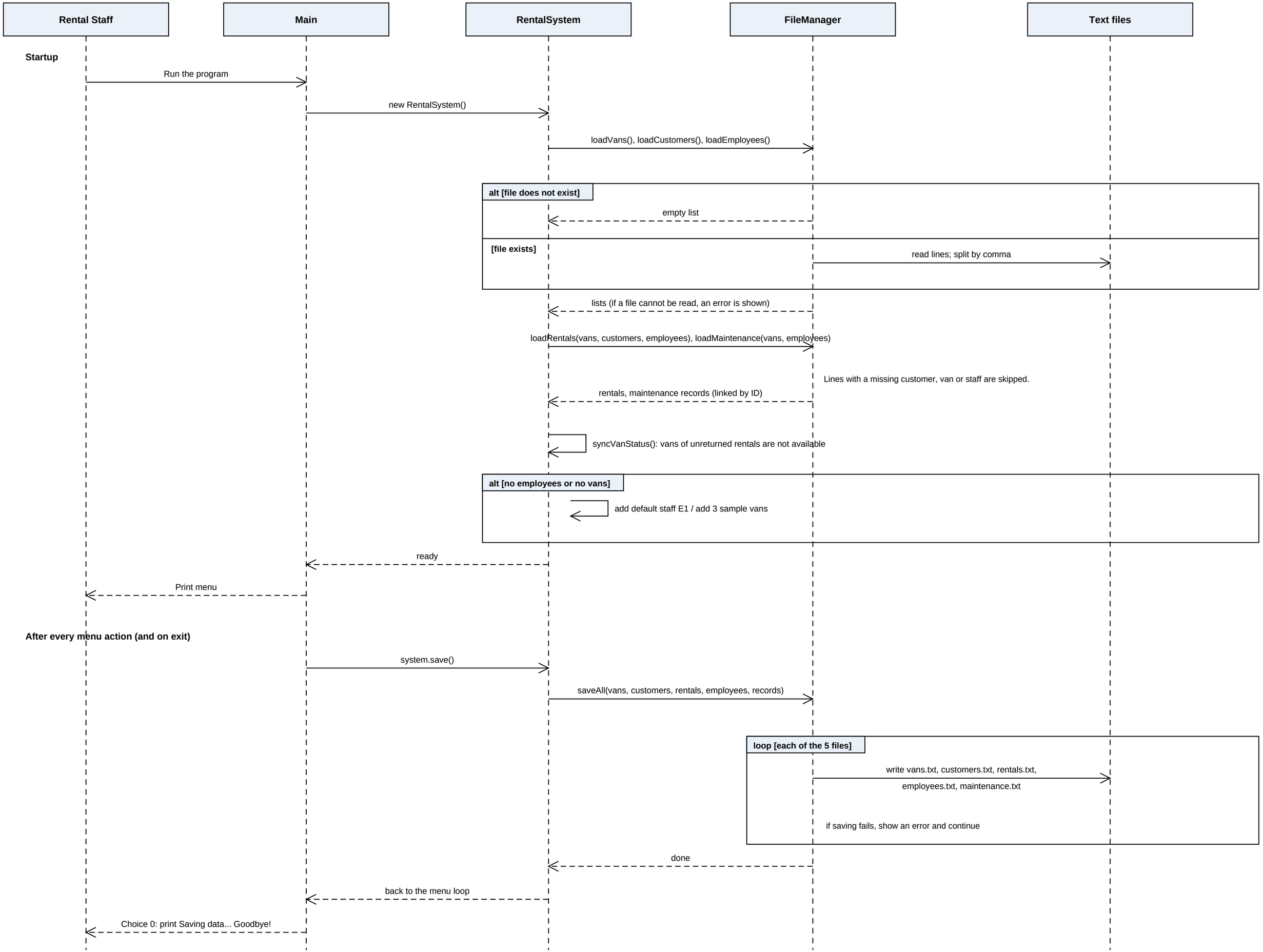
Menu option 10. Late fee is P250 for each day after the end date.



Grand total = rental fee + late fee. The van is Available again.
Then the main loop calls system.save().

8 Load and Save Sequence Diagram

At startup the 5 text files are loaded. After each menu action the data is saved again.



The email and SMS classes write to notifications_log.txt when they send.